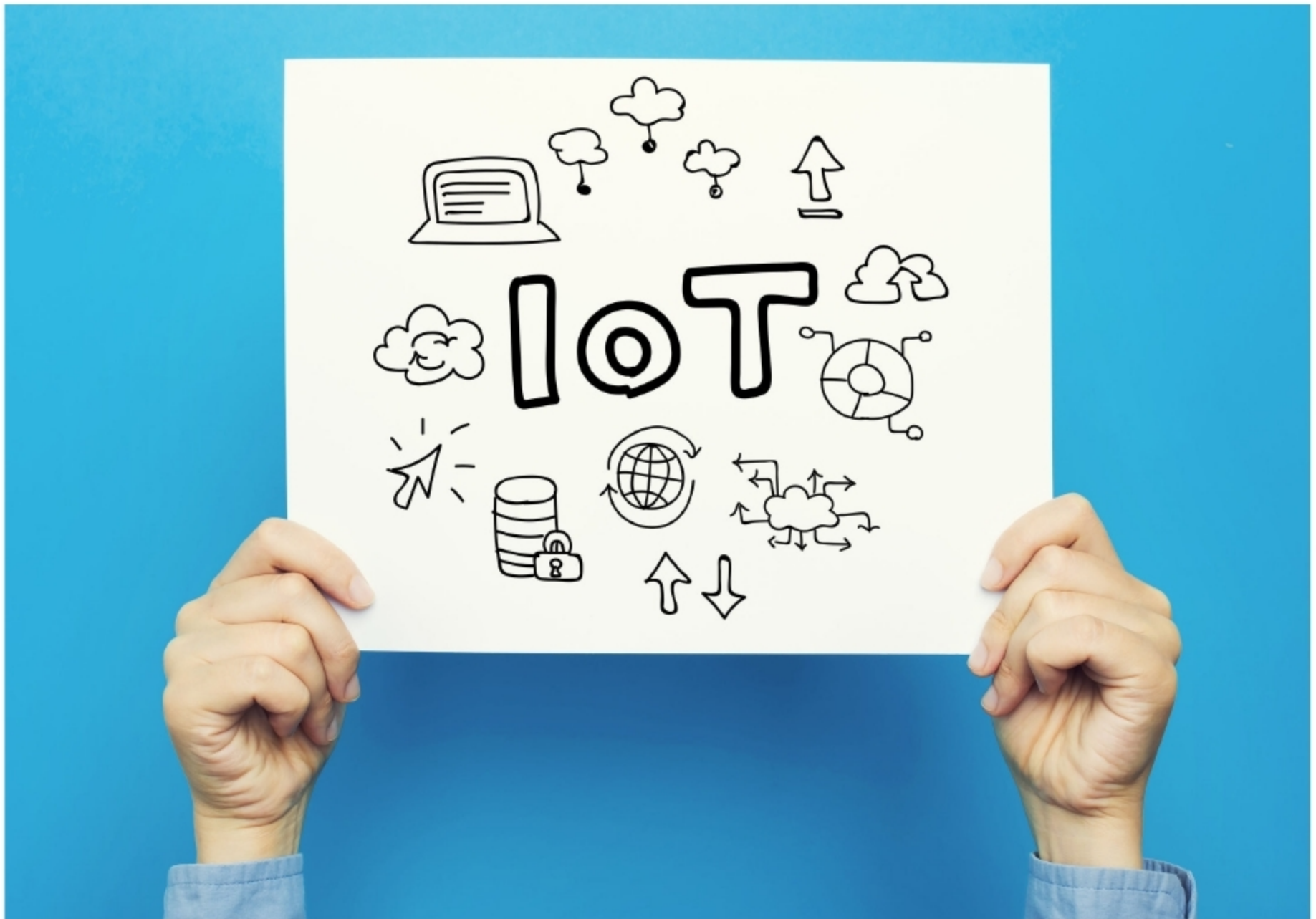


The Top Open Source IoT Platforms for Developers

This article takes a quick look at nine top IoT platforms for developing IoT projects, and discusses their features and properties.



The term ‘the Internet of Things’ was first used by Kevin Ashton in 1999 in the context of supply chain management with radio frequency identification (RFID)-tagged or barcoded items (things) that offered greater efficiency and accountability to businesses.

In an interview, Kevin Ashton told the *RFID Journal*, “If we had computers that knew everything there was to know about things—using data they gathered without any help from us—we would be able to track and count everything and greatly reduce waste, loss and cost. We would know when things needed replacing, repairing or recalling and whether they were fresh or past their best.”

The Internet of Things is at the centre of overlapping Internet-oriented (middleware), things-oriented (sensors) and semantic-oriented (knowledge) visions. The Internet-oriented vision emphasises the networking paradigm and exploits the established IP-based networking infrastructure, in order

to attain efficient connections between devices. It also focuses on developing lightweight protocols to meet IoT specifications. Being things-oriented focuses on physical objects, and on finding the means to identify and integrate them with the virtual world. The semantic-oriented vision aims to utilise semantic technologies, making sense of objects and their data to represent, store, interconnect and manage enormous amounts of information provided by the increasing number of IoT objects.

The IoT can be defined as, “A world of interconnected things that are capable of sensing, actuating and communicating among themselves and with the environment (i.e., smart things or smart objects), while providing the ability to share information and act in parts autonomously to real/physical world events and by triggering processes and creating services with or without direct human intervention.”